



Module 01: SYMPHONY Live City Twin

Lectures Industry briefing · Amit Dua · First Semester 2025–26

1.1 Purpose

SYMPHONY is a hierarchical multi-agent system for adaptive traffic signal control with temporal prediction (Indian Patent Application 202611024014, amended claims 1–10, FER response 31 August 2026). The public industry build must do two jobs at once: show the filed architecture, and run as a live twin for any city an operator can grant access to.

This note describes the live-city system as implemented for the evening briefing. The six-junction Outer Ring Road simulation remains available as the paper demo. It is no longer the only path through the command centre.

Definition 1.1.1 (Live city). A city is live when the operator can select it, the map and timezone change, at least one speed or weather adapter returns a timestamped reading, fusion keeps a rolling window of eight snapshots, and the twin continues if a source fails.

Warning

GitHub Pages never holds Google or TomTom keys. Observe mode uses public APIs and seed geometry. Recommend mode needs a backend. Actuate is blocked until a licensed controller adapter exists.

1.2 City pack

A city is a pack, not a branch of the UI. The pack names the bounding box, IANA timezone, agencies, one corridor, adapter list, and a seed junction table. OpenStreetMap can replace the seed (geometry source marked seed until then).

Key idea

Adding London must not edit Bengaluru code except the registry index. The command centre reads `listCities()` and `getCity(id)`.

Four packs ship:

- Bengaluru, Asia/Kolkata, ORR Silk Board to Marathahalli, Google+TomTom when keyed, Open-Meteo always.
- London, Europe/London, A501 Marylebone to King's Cross, TfL.
- New York, America/New_York, FDR Brooklyn Bridge to 96th Street, NYC Open Data.
- Hong Kong, Asia/Hong_Kong, Island East Central to Quarry Bay, data.gov.hk / TD.

Example

```
getCity("london").tz is Europe/London. An unknown id throws. Tests lock both facts.
```

1.3 Live adapters and fusion

Each speed adapter returns zero or more readings (junctionId, speedKmh, observedAtIso, source, freshnessS) and never throws past the fusion layer. A missing key returns an empty list.

Fusion (claims 3 and 4) assigns $w = 0.5 \cdot \text{freshness} + 0.3 \cdot \text{accuracy} + 0.2 \cdot \text{consistency}$. A source more than 15 km/h from the median is down-weighted by 0.1 unless every source agrees. The window is eight snapshots.

Implementation

On 01 September 2026 this machine received HTTP 200 from Open-Meteo (Bengaluru precipitation) and from NYC Open Data dataset i4gi-tjb9. TfL /Road/all returned 404 without an app key. The health panel must show that amber or red state rather than invent a speed.

Weather from Open-Meteo is not a speed. Rain millimetres scale the time-of-day prior when a billed traffic API is absent, and the panel labels the prior as such.

1.4 Hierarchy on live series

Claim 1 assigns three distinct language models by decision timescale. The smallest, quantized model sits at each junction and must finish inside 200 ms (claim 7). The corridor model computes offsets from predicted speeds (claim 8). The network model issues directives for incidents, VIP movements, and long-term pattern shifts (claim 9). Forecasts at 5, 15, and 30 minutes are inputs to all three levels.

If every speed adapter fails, junction agents switch to stored local logic and the banner reads that the twin is running on local logic. Partial failure fuses whatever remains.

Across languages

Observe / Recommend / Actuate Observe writes advisories only and is the Pages default. Recommend persists suggestions for a partner who holds keys on a server. Actuate would emit phase commands to SUMO TraCI or a field controller. This build refuses Actuate.

1.5 Cameras

Cameras are optional. A city with cameras: none still counts as live on speeds. When a catalogue exists, parse it the way a Sentinel-style GET /api/ingest is parsed: ids, codec, live flag, RTSP, WHEP,



HLS. The page JSON contains HLS or WHEP or JPEG only. RTSP stays on the capture process, forced over TCP, timed from PTS, reconnected with backoff 2–30 s.

Warning

Do not wget a looping file and call it live. Do not publish into the gateway. Open at most four captures.

1.6 Evidence

The FER reply records delay reduced 65%, mean speed up 17%, queue length down 72%, $p < 0.001$, 15 trials, spatial MAE 3.23 km/h, temporal MAE 6.01 km/h. The repository quick-test table is a shorter harness. Do not quote 352% throughput from the long A/B design as if it were that table.

1.7 How to run the briefing

Open the command centre, select Bengaluru, point at source health and the 30-second cycle. Switch to London and confirm Marble Arch, not Silk Board. Open Cities on the website. State that Actuate is off. Direct licence questions to Yushu Excellence Technologies and Amit Dua, CSIS, BITS Pilani.